Srabonti Deb

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Education

University of Alberta — **Edmonton, Alberta**
Master of Science in Computing Science — *Sep 2026 – Present*
- **Relevant Coursework:** Image and Video Processing, AI in Multimedia, Virtual Reality & Tele-Presence.

Rajshahi University of Engineering & Technology (RUET) — **Rajshahi, Bangladesh**
Bachelor of Science in Computer Science & Engineering (CGPA: 3.43 / 4.00) | First Class. — *2019 – 2024*
- **Thesis Topic:** Advancing Deepfake Detection: Integrating Time Series Processing with Vision Transformer.
  Thesis Supervisor: Tasmia Jannat
- **Relevant Coursework:** Linear Algebra, Differential and Integral Calculus, Data Structures and Algorithms, Object Oriented Programming, Database Management, Applied Statistics & Queuing Theory, Digital Image Processing, Neural Network and Fuzzy System, Artificial Intelligence, Data Mining.

Experiences

Young Learners' Research Lab — **Rajshahi, Bangladesh**
Research Assistant — *Jun 2024 – Jun 2026*
Contributed to projects aimed at improving deep learning models. Key projects include:

Explainable Retrieval-Augmented Generation (RAG) (Link)
- Built an explainable RAG system for low-resource and noisy text, benchmarking lexical, dense and hybrid retrieval across retriever–generator pairings to reduce hallucination.
- Utilized BM25 (lexical keyword matching) and Contriever (dense semantic retrieval) models for retrieving relevant answers.
- Integrated these retrieval pipelines with generator models such as Flan-T5 and Llama-3 Instruct.

Gastrointestinal Disease Classification (Link)
- Developed diffusion-based augmentation strategies to address class imbalance in gastrointestinal disease datasets.
- Utilized preprocessing and enhancement techniques including CLAHE, Gaussian smoothing, MSRCR, and bilateral filtering to suppress compression artifacts.
- Implemented stratified k-fold validation for performance assessment.
- Trained and compared multiple architectures (ConvMixer, ConvNeXt-Tiny, Swin Transformer) for disease classification.

Deepfake Video Detection using InceptionNet (Link)
- Developed a 3D InceptionNet framework for six-class deepfake detection on the FaceForensics++ benchmark.
- Implemented a three-phase conditional transfer-learning strategy to address class imbalance, significantly improving minority class performance.
- Evaluated robustness across five manipulation techniques with comprehensive confusion matrix analysis.

Syncrosis Lab — **Dhaka, Bangladesh**
Junior UI Developer — *Sep 2023 – Dec 2023*
- Developed frontend components and features of web solutions using TypeScript, React & Tailwind CSS.
- Contributed to two major client projects: Inventory Management System (an e-commerce platform) & Trip-Portal (a travel agency website), implementing dynamic search, filtering, modals, live data updates and interactive maps.

Code Studio — **Rajshahi, Bangladesh**
UI/UX Designer — *Jul 2020 – Dec 2020*
- Designed the user interface elements including layout, typography, colors, and icons using Adobe XD & Figma.
- Improved the usability, accessibility, and overall interaction experience of the product.

Publications

- Srabonti Deb, Tasmia Jannat, Mohiuddin Ahmed, “Advancing Deepfake Detection: Integrating Time Series Processing with Vision Transformers”, 27th International Conference on Computer & Information Technology (ICCIT-2024), IEEE Bangladesh, December 2024. (DOI: 10.1109/ICCIT64611.2024.11022577)
- Shanjid Hasan Nishat, Srabonti Deb, Mohiuddin Ahmed, “Enhancing Fitness Movement Recognition with Attention Mechanism and Pre-Trained Feature Extractors”, arXiv:2509.02511.

Projects

Multi-PDF Chatbot | Python, LangChain, FAISS, Hugging Face, Streamlit (Link)

- Built a context-aware chatbot that answers questions across multiple uploaded PDFs using open-source NLP models.
- Utilized Sentence-Transformers (all-MiniLM-L6-v2) and LangChain with FAISS for semantic retrieval and similarity search.
- Integrated Llama-3.1-8B-Instruct with ConversationBufferMemory for multi-turn question answering, with an interactive Streamlit interface.

E-commerce Website | HTML, CSS, Bootstrap, JavaScript, PHP, MySQL (Link)

- Built a responsive storefront with autocomplete search and product previews.
- Implemented hashed credentials and secure payment gateway integration; improved performance and scalability with index-optimized queries.

Save-Me Security App | Java, Android SDK (Link)

- Developed a user database capturing user details and five trusted contact numbers.
- Implemented an emergency feature to send GPS location and request assistance with a single button press.

Skills

- **Programming Languages:** Python, C, C++, Java, MATLAB, HTML/CSS, JavaScript, MySQL, PHP.
- **Libraries/Frameworks:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, React, Bootstrap, Arduino SDK.
- **Tools & Applications:** Git, GitHub, Jupyter Notebook, XAMPP, Overleaf, Adobe XD, Adobe Illustrator, Canva, Cisco Packet Tracer, Logisim.

Leadership Experiences

Class Representative, CSE (Session 18-19), RUET: Managed class schedules, facilitated communication between students and faculty, and coordinated departmental activities.

Awards

- **National Higher Secondary Scholarship:** Full tuition fee scholarship and stipend during undergraduate studies.
- **National Secondary School Scholarship:** Full tuition fee scholarship and stipend during higher secondary studies.

Language

Bengali (Native), English (Fluent)
IELTS: Overall Band Score 7.0 (Listening: 7.0, Reading: 7.5, Writing: 6.5, Speaking: 7.0)

References

1. Prof. Dr. Md. Al Mamun
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